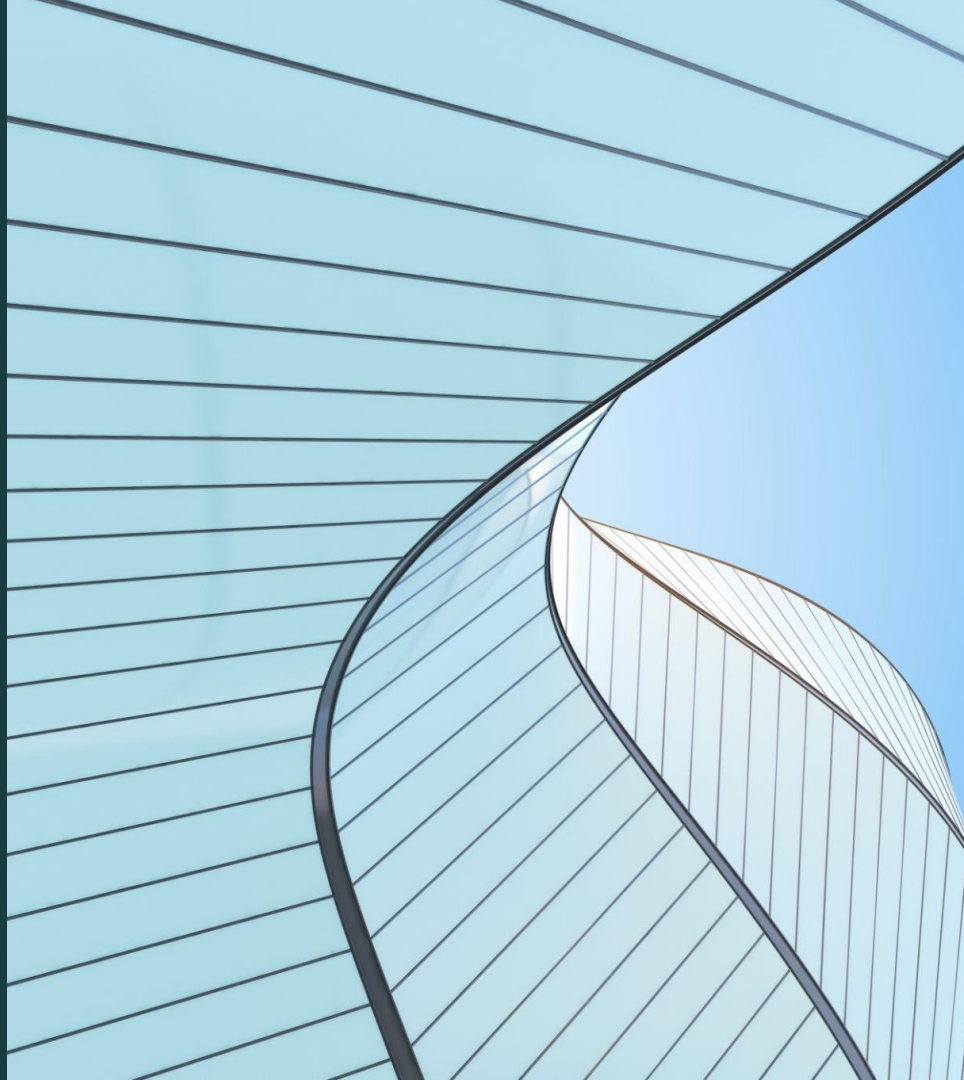
Baseline Model for Group D

Group Members: Arnav K, Alex, Arthur, Emily



Content *overview*

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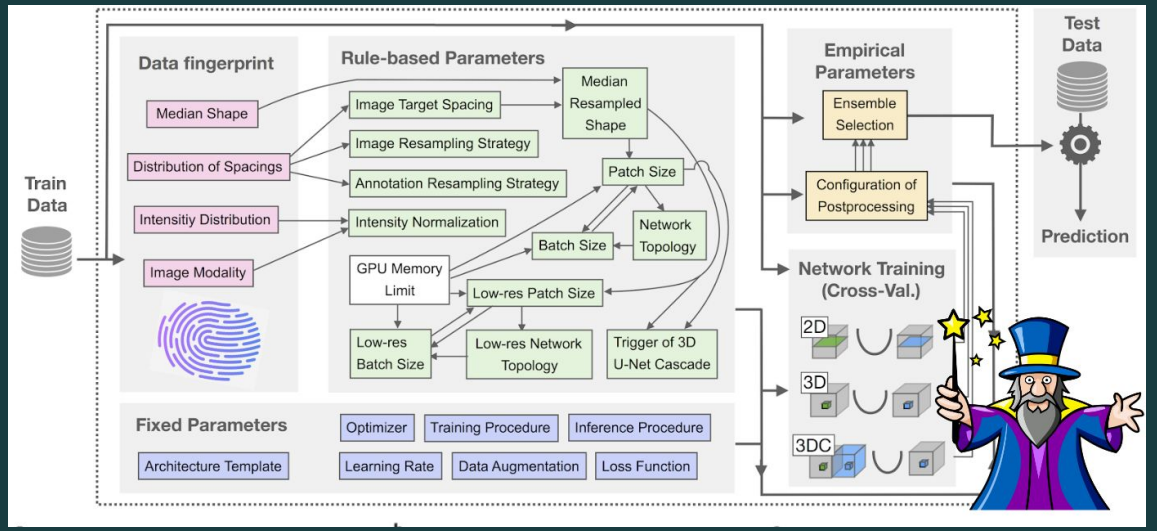
Project and model we will use

- Our project aims to train a model in prostate segmentation at one site and measure how well it generalises to unseen hospitals (unlike SAML which trained on 5 and generalised to 1)
- We will use the same sites as in SAML – 6 sites comprised of 3 datasets: NCI-SCBI (A & B); I2CVB (C); PROMISE12 (D, E, and F)
- The model we will use is the nnU-Net v2 2D segmentation model

Dataset	Institution	Case num	Field strength(T)	Resolution(in/ through plane)(mm)	Endorectal Coil	Manufactor
Site A	RUNMC	30	3	0.6-0.625/3.6-4	Surface	Siemens
Site B	BMC	30	1.5	0.4/3	Endorectal	Philips
Site C	HCRUDB	19	3	0.67-0.79/1.25	No	Siemens
Site D	UCL	13	1.5 and 3	0.325-0.625/3-3.6	No	Siemens
Site E	BIDMC	12	3	0.25/2.2-3	Endorectal	GE
Site F	HK	12	1.5	0.625/3.6	Endorectal	Siemens

Overview of the 2D nnU-Net model

- Data Fingerprint (Data Extraction)
- Rule-Based Parameters (System Self-Design)
- Fixed Parameters (Hard Coded Logic)
- Empirical Parameters (Post-Training Optimisation)
- Network Training & Prediction (Execution)



Site used to train the model

- For training we will use site A (from the NCI-SCBI dataset):
- Has n = 30 (largest number of cases in SAML)
- High field strength (3T)
- Surface coil used (only site that uses one)
- Sites A & B are from the same dataset but have different manufacturers and different endorectal coils
- Sites C, D, E, & F are from different datasets than site A and sites E & F are endorectal coils

Dataset	Institution	Case num	Field strength(T)	Resolution(in/ through plane)(mm)	Endorectal Coil	Manufactor
Site A	RUNMC	30	3	0.6-0.625/3.6-4	Surface	Siemens
Site B	BMC	30	1.5	0.4/3	Endorectal	Philips

Modification to models

- We have to modify the voxel spacing & patch sizing for fixing the physical scale alignment & pixel layout, respectively
- We have to do intensity domain modifications for the different coils used (surface & endorectal) and the different field strengths (3T & 1.5T).
- Removing the batch normalisation and replacing it with group normalisation (so we can train on one dataset)
- Performance evaluation tests; we will use SAML loss functions

Questions?

Resources

https://github.com/mic-dkfz/nnunet

https://liuquande.github.io/SAML/

https://www.sciencedirect.com/org/science/article/pii/S1195947925000394